EXPERIMENT 3: TOKENIZATION USING BERT AND GPT TOKENIZERS

Aim

To perform tokenization using BERT and GPT tokenizers and compare their token outputs and token IDs.

Algorithm

1. Import AutoTokenizer from Transformers.
2. Load the BERT tokenizer and GPT-2 tokenizer.
3. Provide a common input sentence.
4. Tokenize the sentence using both tokenizers.
5. Generate the corresponding token IDs.
6. Display and compare the outputs.

Program

```
# Experiment 3: Compare BERT and GPT Tokenizers

from transformers import AutoTokenizer

# Load tokenizers

bert_tokenizer = AutoTokenizer.from_pretrained(

    "bert-base-uncased"

)

gpt_tokenizer = AutoTokenizer.from_pretrained(

    "gpt2"

)
```

```
# Input sentence

text = "Generative AI is changing the world."

# BERT tokenization

bert_tokens = bert_tokenizer.tokenize(text)

bert_ids = bert_tokenizer.encode(text)

# GPT tokenization

gpt_tokens = gpt_tokenizer.tokenize(text)

gpt_ids = gpt_tokenizer.encode(text)

# Display results

print("Input Text:", text)

print("\nBERT Tokens:")

print(bert_tokens)

print("\nBERT Token IDs:")

print(bert_ids)

print("\nGPT Tokens:")

print(gpt_tokens)

print("\nGPT Token IDs:")

print(gpt_ids)

# Compare token counts

print("\nBERT Token Count:", len(bert_tokens))

print("GPT Token Count:", len(gpt_tokens))
```

Input

Generative AI is changing the world.

Expected Output

The token strings below illustrate the typical result; the program will display the exact token IDs.

Input Text: Generative AI is changing the world.

BERT Tokens:

['generative', 'ai', 'is', 'changing', 'the', 'world', '.']

BERT Token IDs:

[101, ..., 102]

GPT Tokens:

['Gener', 'ative', 'ĠAI', 'Ġis', 'Ġchanging', 'Ġthe', 'Ġworld', '.']

GPT Token IDs:

[...]

BERT Token Count: 7

GPT Token Count: 8

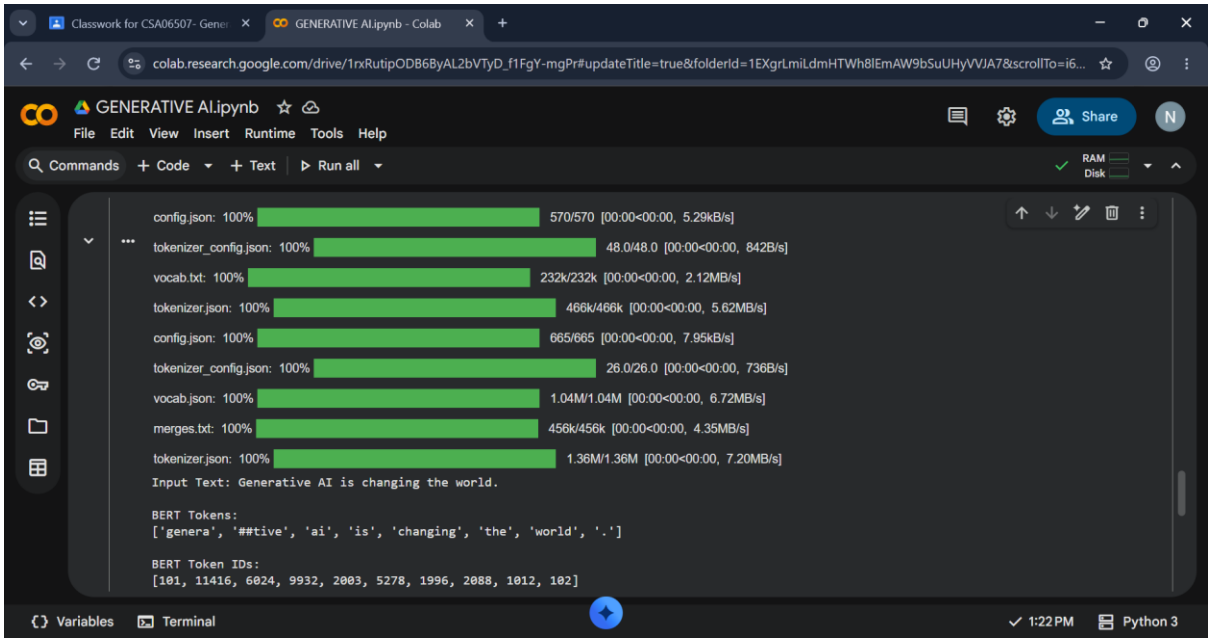
Important: The BERT IDs above use an ellipsis to indicate omitted values, not literal program output. Run the code to obtain the complete IDs.

Comparison

Feature	**BERT**	**GPT-2**
Tokenization method	WordPiece	Byte-level BPE

Capitalization	bert-base-uncased converts text to lowercase	Generally preserves capitalization
Space representation	Spaces are not usually displayed as separate token markers	Ġ commonly indicates a preceding space
Common applications	Text understanding and classification	Text generation

The number of tokens depends on the input sentence and tokenizer.



Result

Thus, the input sentence was tokenized using BERT and GPT-2 tokenizers, and their token strings, token IDs, and token counts were compared.